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Linear differential equations.

A differential equation of the form $\frac{dy}{dx} + py = Q$ is called linear differential equation where P and Q are functions of x or constant.

To solve this type of DE, first we have to find out I.F = $e^{\int P dx}$

Then solution will be

$$y \cdot I.F = \int Q \cdot I.F dx + C$$

$$\# (1-x^2) \frac{dy}{dx} - xy = 1$$

$$\Rightarrow \frac{dy}{dx} - \frac{x}{1-x^2} y = \frac{1}{1-x^2}$$

$$\text{Now, I.F} = e^{\int -\frac{x}{1-x^2} dx} = e^{-\frac{1}{2} \ln(1-x^2)} = \sqrt{1-x^2}$$

$\therefore$ Solution of the given equation is

$$\begin{aligned} y \cdot \sqrt{1-x^2} &= \int \frac{1}{1-x^2} \cdot \sqrt{1-x^2} dx + C \\ &= \int \frac{1}{\sqrt{1-x^2}} dx + C \\ &= \sin^{-1} x + C \end{aligned}$$

$$\Rightarrow y = \frac{\sin^{-1} x}{\sqrt{1-x^2}} + \frac{C}{\sqrt{1-x^2}}$$

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$$(1+x^2) \frac{dy}{dx} + y = \tan^{-1} x$$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{1+x^2} y = \frac{\tan^{-1} x}{1+x^2}$$

$$\tan^{-1} x = z$$

$$\frac{1}{1+x^2} dx = dz$$

$$\therefore I.F. = e^{\int \frac{1}{1+x^2} dx} = e^{\tan^{-1} x}$$

$$\text{Soln: } y e^{\tan^{-1} x} = \int \frac{\tan^{-1} x e^{\tan^{-1} x}}{1+x^2} dx + c$$

$$= \int z e^z dz + c$$

$$= (z-1) e^z + c$$

$$= (\tan^{-1} x - 1) e^{\tan^{-1} x} + c$$

$$\Rightarrow y = \frac{\tan^{-1} x - 1}{e^{\tan^{-1} x}} \neq e / e^{\tan^{-1} x}$$

$$= (\tan^{-1} x - 1) e^{-\tan^{-1} x} + e^{-\tan^{-1} x}$$

$$\# (2+y^2) dx = (yx + 2y + y^3) dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{xy + 2y + y^3}{2+y^2} dy$$

$$\Rightarrow \frac{dx}{dy} - \frac{xy}{2+y^2} = \frac{y(2+y^2)}{2+y^2}$$

$$\Rightarrow \frac{dx}{dy} - \frac{y}{2+y^2} x = y$$

$$\therefore I.F. = e^{\int \frac{-y}{2+y^2} dy} = e^{-\frac{1}{2} \ln(2+y^2)} = \frac{1}{\sqrt{2+y^2}}$$

$$\text{Soln: } x / \sqrt{2+y^2} = \int \frac{y}{\sqrt{2+y^2}} dy + c$$

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$$\Rightarrow \frac{x}{\sqrt{2+y^2}} = \frac{1}{2} \cdot 2 \cdot \sqrt{2+y^2} + C$$

$$\Rightarrow x = (2+y^2) + C(\sqrt{2+y^2})$$

$$\# (1+y^2) dx = (\tan^{-1}y - x) dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{\tan^{-1}y}{1+y^2} - \frac{x}{1+y^2}$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1}y}{1+y^2}$$

$$\therefore I.F = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1}y}$$

$$\therefore \text{Soln: } x e^{\tan^{-1}y} = \int \frac{\tan^{-1}y e^{\tan^{-1}y}}{1+y^2} dy + C$$

$$\Rightarrow x e^{\tan^{-1}y} = (\tan^{-1}y - 1) e^{\tan^{-1}y} + C$$

$$\Rightarrow x = (\tan^{-1}y - 1) + C e^{-\tan^{-1}y}$$

$$\# (1+x^2) \frac{dy}{dx} + 2xy = 4x^2$$

$$\# \frac{dy}{dx} + 2xy = 2x$$

$$\# \frac{dy}{dx} + \left(\frac{1-2x}{x^2} \right) y = 2$$

$$\begin{aligned} \therefore I.F &= e^{\int \frac{1-2x}{x^2} dx} = e^{\int \left(\frac{1}{x^2} - \frac{2x}{x^2} \right) dx} \\ &= e^{\left(-\frac{1}{x} - \ln x^2 \right)} \\ &= e^{-\frac{1}{x}} e^{\ln \frac{1}{x^2}} \\ &= \frac{e^{-\frac{1}{x}}}{x^2} \end{aligned}$$

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$$\therefore \text{sol}^n: y \frac{e^{-\frac{1}{x}}}{x^2} = \int \frac{e^{-\frac{1}{x}}}{x^2} dx + c$$
$$= e^{-\frac{1}{x}} + c$$

$$\Rightarrow y = x^2 + c e^{\frac{1}{x}} \quad \underline{\underline{A_2}}$$